

This analysis uses IT salary data collected from professionals working in Germany, spanning ages 20 to 69, with a focus on understanding compensation dynamics. Key attributes include total experience, years working specifically in Germany, programming skills, and company size. This report integrates data cleaning, exploratory visualization, and comparative analysis across roles, experience levels, and industry sectors to reveal patterns influencing salary growth in the technology sector.

The dataset reveals that the respondents are predominantly young to mid-career professionals, with ages spanning from 20 to 69 years and an average age of approximately 32.5 years. This age distribution suggests a workforce with substantial potential for career growth, with insights on skill development needs and potential future earnings. The dataset includes detailed experience metrics, capturing both total professional experience and specific years of experience within Germany. This segmentation allows for a nuanced analysis of career progression and geographic work patterns, potentially highlighting the mobility of IT professionals and their adaptation to the German labor market. The YearlySalaryInEUR field indicates considerable variance in compensation levels, likely influenced by factors such as seniority, technical expertise, and geographic location. The presence of some extreme salary values suggests the presence of high-earning specialists or executives.

Data Cleaning and Preparation

a/ Upon reviewing the merged dataset, several data quality issues were identified that could impact analysis accuracy and reliability. Missing values were found in key fields, such as Age, YearlySalaryInEUR, NumberOfVacationDays, and AnnualSalaryOneYearAgo, potentially limiting the reliability of analyses on salary and benefits. Some numerical fields, like TotalYearsOfExperience and YearsOfExperienceInGermany, were stored as non-numeric data, requiring conversion for accurate calculations. Additionally, several fields showed extreme values (e.g., extremely high salaries) that may represent outliers and could skew results. Inconsistencies in text formatting, including varying cases and leading/trailing whitespace in categorical fields like Gender and Position, could lead to errors when grouping or aggregating data. Duplicate entries, particularly in the RecordID field, were also found, potentially distorting analysis by over-representing certain records. Addressing these issues was essential to prepare the data for a reliable and robust analysis.

b/ To address the identified data quality issues, several cleaning steps were applied to ensure data integrity and consistency. First, all character fields were standardized to lowercase, and whitespace was trimmed to maintain uniformity and prevent duplication due

to case or spacing differences. Duplicate records were removed based on the RecordID field, retaining only unique entries to avoid over-representation of certain records. To handle data type inconsistencies, numerical fields like TotalYearsOfExperience and YearsOfExperienceInGermany were converted from character to numeric, allowing accurate calculations. Missing values in key numeric fields, including YearlySalaryInEUR and AnnualSalaryOneYearAgo, were replaced with the median value of each respective column. This approach minimized the potential impact of outliers while ensuring data completeness. Finally, all categorical data were checked for inconsistencies, and naming conventions were standardized. These steps created a cleaned, reliable dataset, providing a robust foundation for further analysis.

Salary Distributions by Job Role and Year

In both 2019 and 2020, salary distributions across job roles showed significant variability, influenced by specialization and market demand. By 2020, salary ranges became narrower, with most roles having more defined pay structures. However, Sales and Agile Coaching showed only median values, indicating that many salaries in these roles were outliers, contributing to greater variability. The lowest-paid roles in 2020 were Design and Quality Assurance.

In 2019, salary ranges were wider, reflecting more variability across roles. Agile & Coaching and Research had broad salary ranges and high medians, indicating high earning potential due to specialized skills and leadership. Design and Quality Assurance roles had lower medians and narrower salary ranges, suggesting more standardized pay structures, but outliers showed that higher earnings were possible for those with advanced expertise. Software Development had a moderate median salary with a wide range, where experience and technical skills influenced compensation. The lowest-paid roles in 2019 were Recruiting and Consulting & Support.

Over 2 years, while salary ranges and medians shifted between 2019 and 2020, high-paying roles like Agile & Coaching and Research continued to command premium salaries. Furthermore, salaries increased modestly from 2019 to 2020, with specialized and leadership roles seeing the most significant growth.

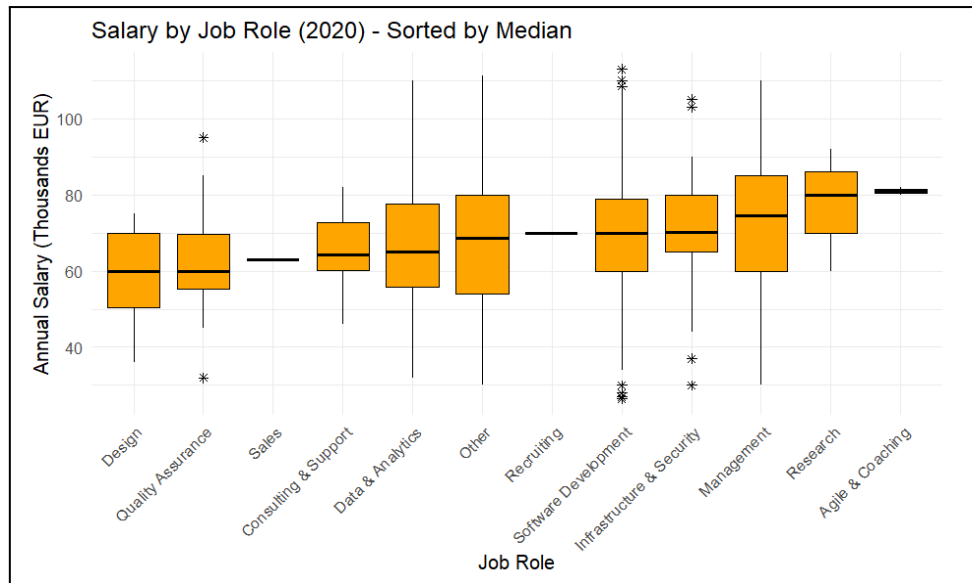


Figure 1. The box plot showing salary ranges in 2019 based on Job Role

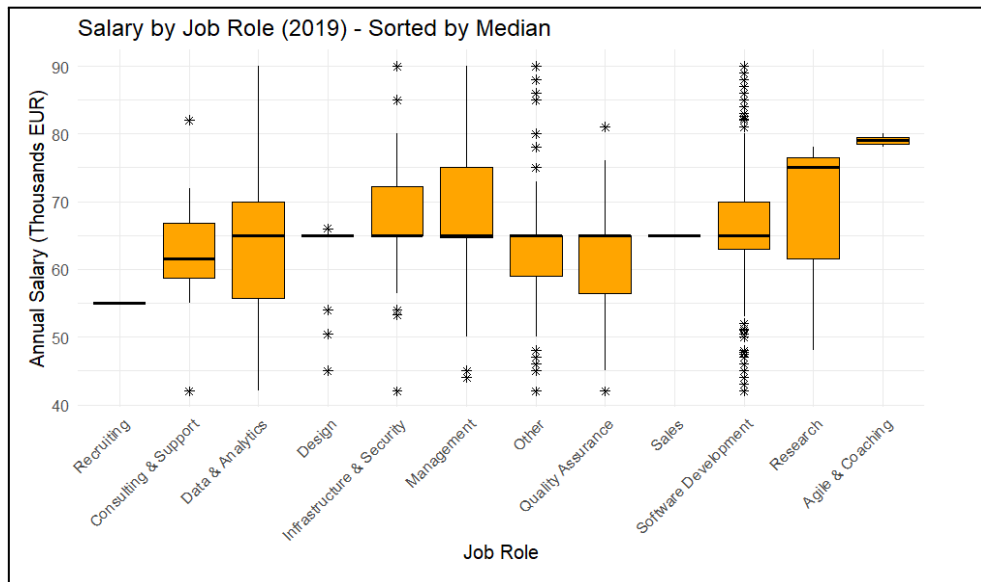


Figure 2. The box plot showing salary ranges in 2019 based on Job Role

JobCategory <chr>	Median_Salary_2020 <dbl>	Median_Salary_2019 <dbl>
Design	60000	65000
Quality Assurance	60000	65000
Sales	63000	65000
Consulting & Support	64250	61500
Data & Analytics	65000	65000
Other	68500	65000
Recruiting	70000	55000
Software Development	70000	65000
Infrastructure & Security	70200	65000
Management	74500	65000

Research	80000	75000
Agile & Coaching	81000	79000

Figure 3. The table showing the difference between the median salary in 2019 and 2020 based on Job Role

Experience vs Salary: Career Growth Insights

The analysis of salary trends in the IT industry shows that years of experience significantly influence salary levels, though the impact varies by job role and experience range. For new entrants, salaries are relatively stable around €50,000 per year, with limited outliers. As professionals gain experience, particularly between 1-3 years and 5-10 years, the salary range expands, with upper outliers indicating potential for higher earnings. However, from 10-20 years of experience, a shift occurs: lower outliers emerge, suggesting that some individuals, despite their extensive experience, may not be keeping up with market trends due to skill gaps or shifts in the industry. Interestingly, for those with 20+ years of experience, the median salary remains relatively stable, but the range becomes narrower, implying that seniority does not always lead to significantly higher pay.

Job category analysis reveals that roles in Data & Analytics exhibit the most significant salary growth, with entry-level salaries around €50,000 and senior roles reaching up to €110,000. Software Development shows a steady increase across experience levels, while Infrastructure & Security and Quality Assurance demonstrate more moderate growth patterns. Management roles tend to plateau earlier compared to technical positions.

An unexpected trend is that some professionals with 20+ years of experience, particularly in non-technical roles, earn less than their mid-career counterparts. In contrast, technical roles, especially in Data & Analytics and Software Development, continue to see salary growth even at senior levels, highlighting the strong demand for specialized skills.

These findings indicate that while experience generally drives salary growth, its impact varies significantly by job category, with technical roles showing the strongest correlation between experience and compensation.

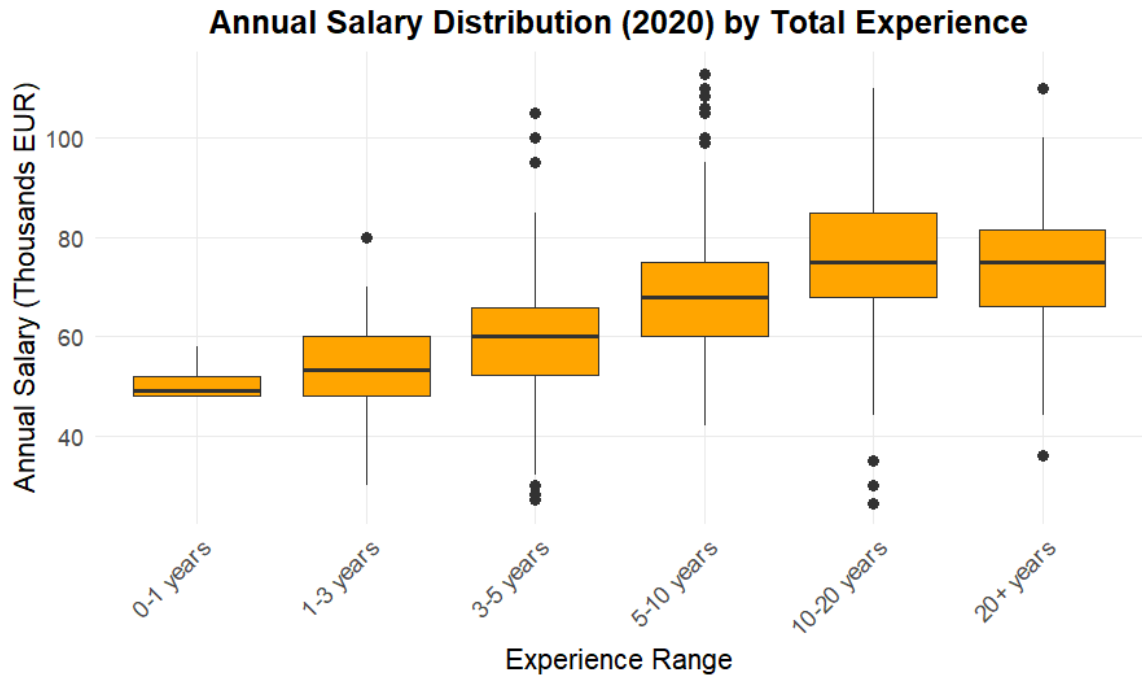


Figure 4. The box plot shows the salary ranges based on total years of experience

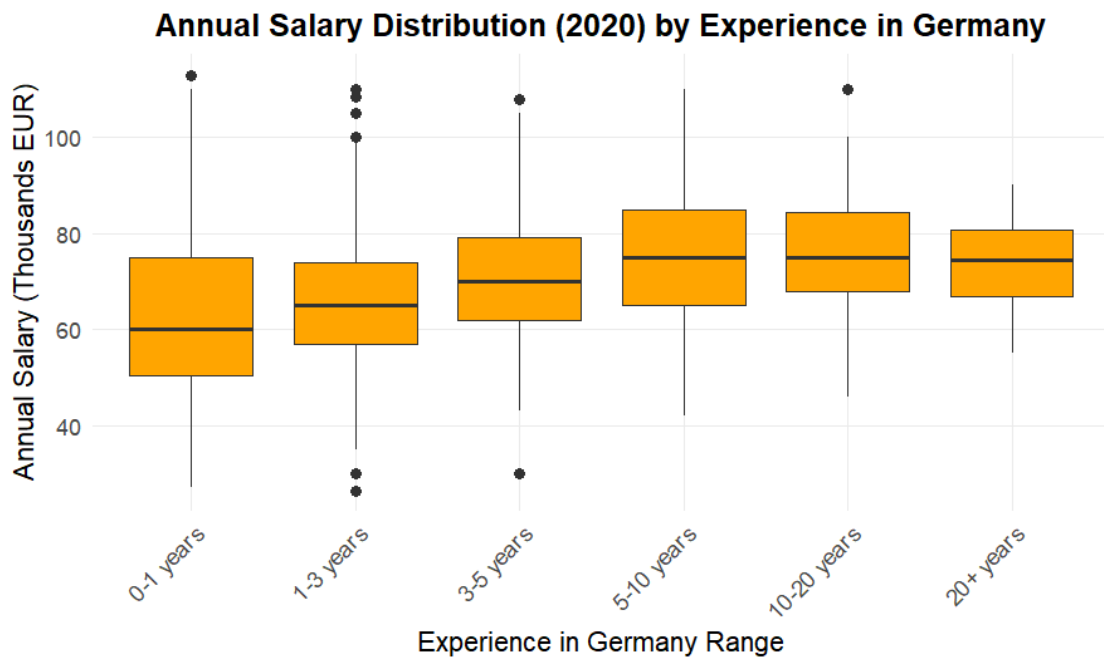


Figure 5. The box plot shows the salary ranges based on years of experience in Germany

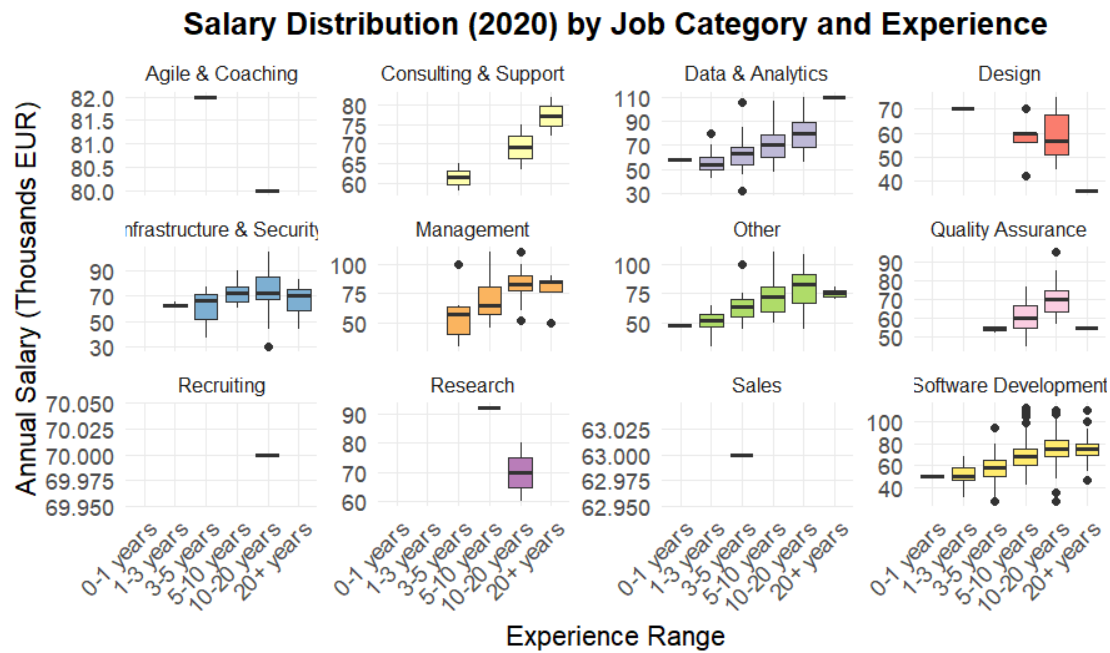


Figure 6. The box plot shows the salary distribution based on Job Category and experience year

Impact of Skills and Company Characteristics on Salary

In addition to experience and job role, factors such as **Main Language**, **Main Programming Language**, **Company Type**, and **Company Size** significantly shape IT salaries by influencing market demand, skill specialization, and available resources.

Main Language is a key driver of salary, as multilingual professionals, particularly those fluent in **English** and **French**, are in high demand for global and multinational firms (Organisation for Economic Co-operation and Development, 2023). These language skills can command higher compensation, as they enable companies to tap into international markets and collaborate with diverse teams.

Similarly, **Main Programming Language** expertise strongly impacts earning potential. In-demand languages like **Go**, **Python**, and **TypeScript** are critical in emerging technologies, often yielding higher salaries. For instance, Go developers earn a median of €75,000, as the language is associated with high-performance computing and cloud infrastructure (Anderson, 2022). While modern languages offer higher pay, traditional languages like **Java** and **C++** maintain steady demand due to their application in established enterprise systems, ensuring competitive compensation. **SQL** also remains a valuable skill, with consistent demand across sectors, albeit at lower salary medians.

The **Company Type** also plays a significant role. Fintech companies and Technology Startups offer the highest salaries, with median compensation around €75,000, due to their reliance on specialized skills and high profit margins (McKinsey & Company, 2023). In contrast, consulting and services roles tend to offer more stable but lower salaries.

Company Size is another crucial determinant. Larger organizations (over 1,000 employees) offer the highest median salaries, around €72,000, along with better benefits and career progression opportunities. Medium-sized companies offer competitive pay but with less variation, while smaller firms often present more unpredictable salary structures, reflecting their limited resources and varying business models.

For newcomers entering the IT field, strategic career planning is essential. Starting with a large corporation can offer structured training programs and stable career paths while gaining experience in **fintech** or **tech** sectors can provide high earning potential for those seeking challenging and dynamic environments. Developing a versatile skill set, particularly in modern programming languages like Go and TypeScript, alongside foundational knowledge in languages like Java and SQL, enhances employability and opens doors to higher-paying roles.

Progressing from large companies to startups after building core skills offers exposure to diverse work environments and opportunities to expand professional networks. Moreover, focusing on companies with structured graduate programs or strong mentorship can accelerate career growth. By understanding the intersection of language proficiency, programming expertise, company type, and size, newcomers can make informed decisions that align with both immediate earning potential and long-term career success in the competitive IT market.

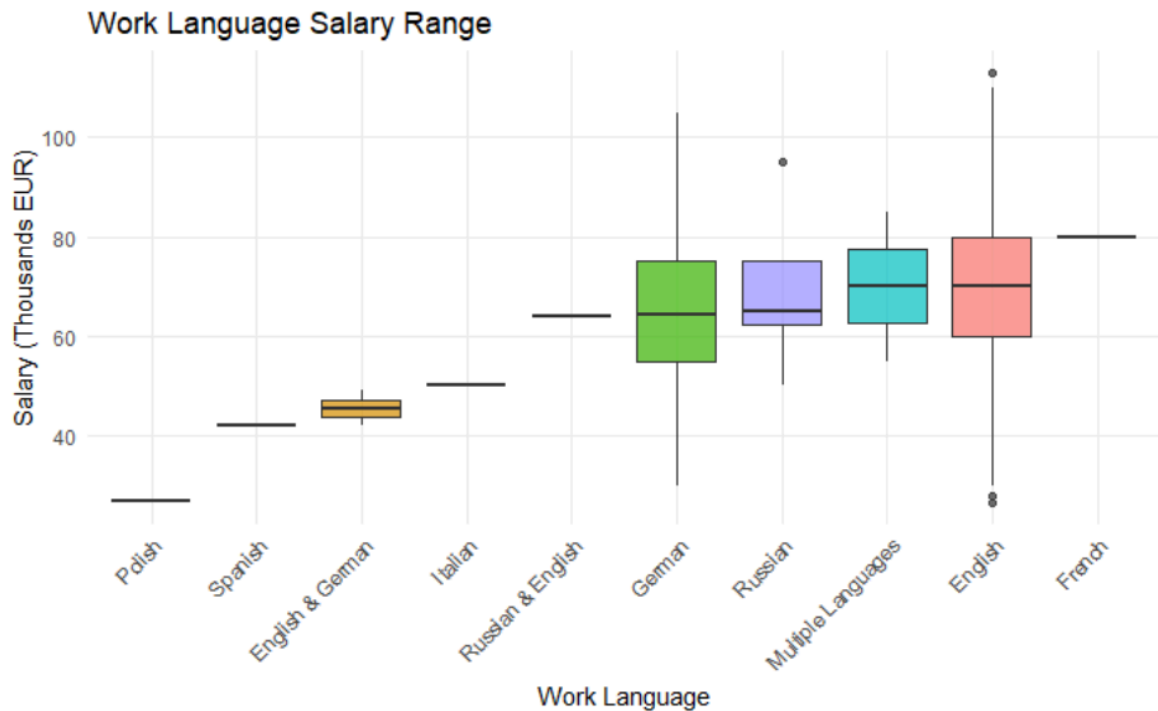


Figure 7. The box plot shows the salary distribution based on Main Language speaking at work

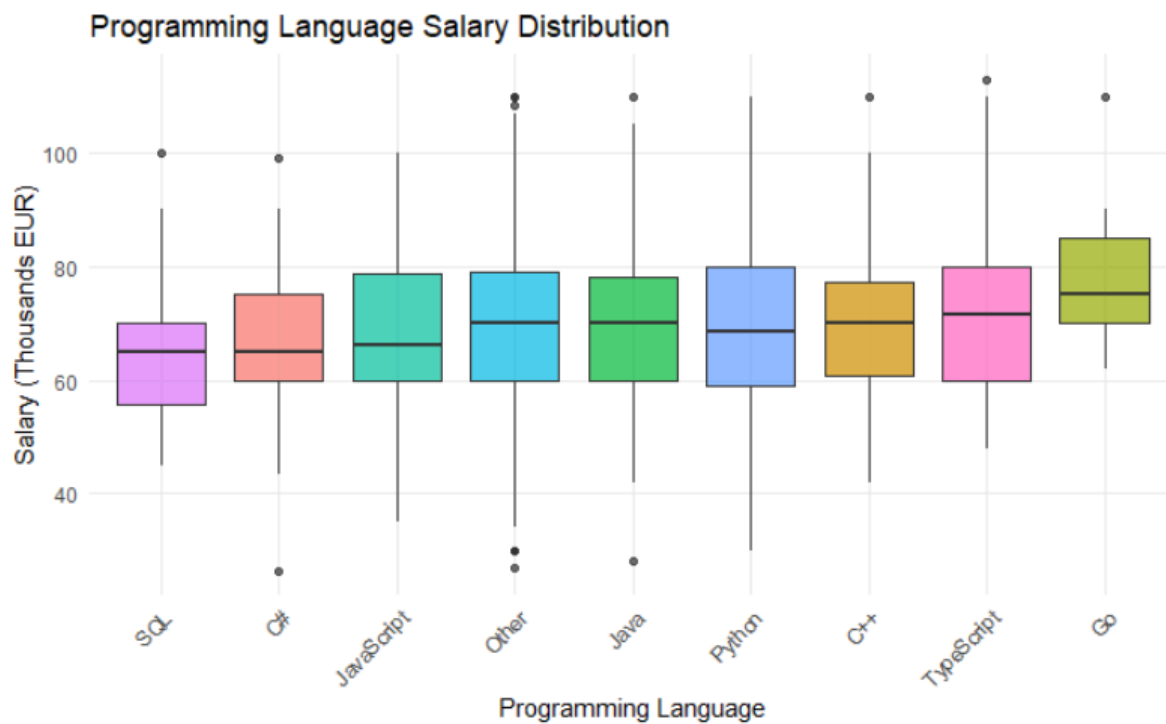


Figure 8. The box plot shows the salary distribution based on Main programming Language

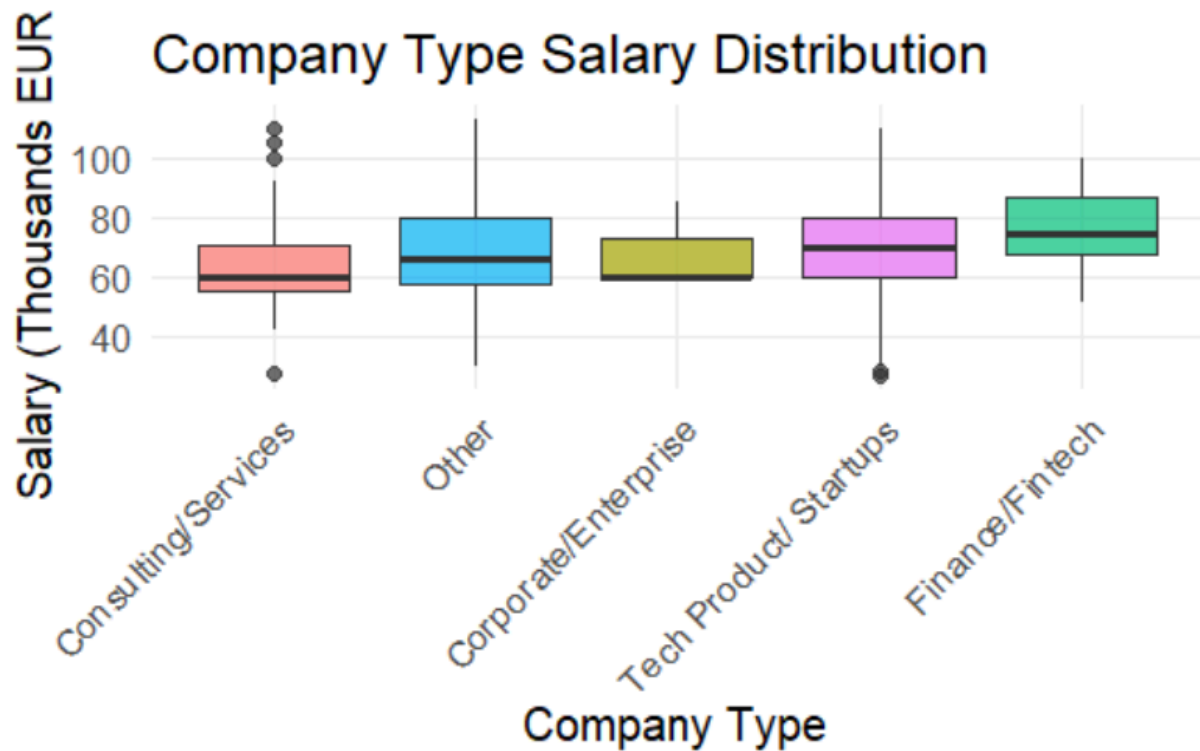


Figure 9. The box plot shows the salary distribution based on Company Type

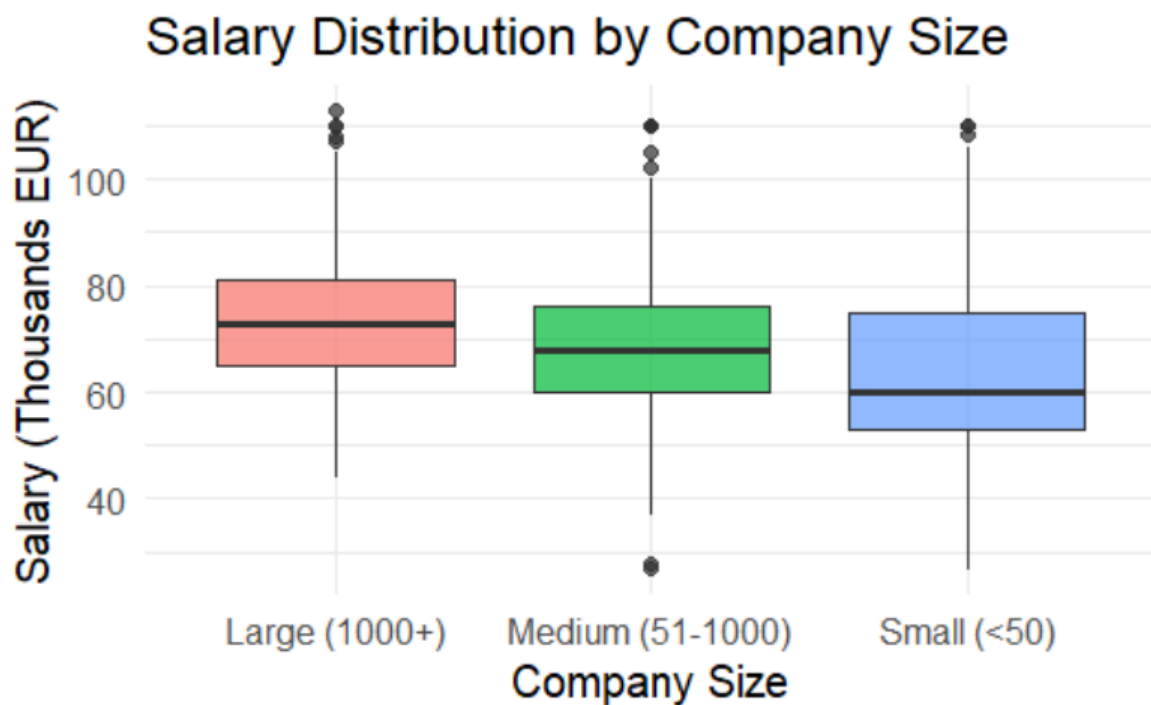


Figure 10. The box plot shows the salary distribution based on Company Size

Reference list:

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